

Cybersecurity Internship: Phase-2 Final Report

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Executive Summary

This report details the execution and completion of Phase-2 (Weeks 4-6) of the DevelopersHub Corporation cybersecurity internship program. Building upon foundational security measures, this phase focused on implementing advanced threat detection mechanisms, conducting ethical hacking to exploit and remediate vulnerabilities, and performing comprehensive security audits to ensure secure application deployment.

Week 4: Advanced Threat Detection & Web Security Enhancements

The objective of Week 4 was to establish proactive defense systems and harden the application's API endpoints against unauthorized access and abuse.

Implementations:

- **Intrusion Detection Monitoring:** Configured `Fail2Ban` to monitor server logs in real-time. Created custom jail rules to issue temporary IP bans after multiple failed login attempts, mitigating brute-force risks.
- **API Security & Rate Limiting:** Integrated the `express-rate-limit` middleware to cap API requests, effectively preventing denial-of-service (DoS) and brute-force attacks. Cross-Origin Resource Sharing (CORS) was strictly configured to whitelist only trusted domains.
- **Header Hardening:** Implemented a robust Content Security Policy (CSP) to block unauthorized external scripts. Enforced HTTP Strict-Transport-Security (HSTS) headers to guarantee secure, encrypted client-server communication.

Week 5: Ethical Hacking & Exploiting Vulnerabilities

Week 5 transitioned into an offensive security mindset. Using industry-standard penetration testing toolkits, I actively scouted the application for exploitable logic flaws and patched them.

Methodology & Remediation:

- **Reconnaissance & SQLi Exploitation:** Utilized Kali Linux and `SQLMap` to test data inputs. Identified a vulnerable parameter allowing SQL injection (SQLi).
- **SQL Injection Remediation:** Refactored the backend database queries to exclusively use prepared statements, effectively neutralizing the SQLi vulnerability by separating code from user-supplied data.
- **CSRF Vulnerability Testing:** Leveraged Burp Suite to intercept requests and successfully executed a Cross-Site Request Forgery (CSRF) attack against the session state.
- **CSRF Protection:** Implemented the `csrf` middleware within the Node.js application. All state-changing operations now require a valid, unique CSRF token, verified during Burp Suite re-testing.

Week 6: Advanced Security Audits & Final Deployment

The final week focused on validating the implemented security controls against industry standards and preparing the application for a secure production release.

Auditing & Deployment Steps:

- **Comprehensive Security Audits:** Conducted extensive automated and manual audits using OWASP ZAP, Nikto, and Lynis. Ensured total compliance with OWASP Top 10 best practices.
- **Secure Docker Deployment:** Containerized the application using Docker. Implemented security best practices by running containers as non-root users, minimizing the base image size, and scanning images for known vulnerabilities.
- **Final Penetration Test:** Executed a final, comprehensive penetration test utilizing Burp Suite and Metasploit. The results confirmed that all previously identified vulnerabilities (XSS, SQLi, CSRF, and misconfigurations) were successfully remediated.

Bonus Challenge: Security Excellence

To exceed baseline requirements, additional advanced security protocols were integrated into the deployment pipeline.

- **Zero Trust Architecture & WAF:** Adopted Zero Trust principles by enforcing strict, least-privilege resource access. Additionally, deployed a Web Application Firewall (WAF) layer to filter and inspect incoming malicious traffic patterns.

Conclusion: The application has successfully passed all Phase-2 security audits and penetration tests. The source code, configurations, and corresponding documentation have been pushed to the GitHub repository for final review.